PFIZER DIGITAL & TECHNOLOGY

HACKATHON 2026

Technical Resources

Introduction

This document contains questions and resources to guide your brainstorming about the problem statements for the Pfizer Digital & Technology Hackathon 2026.

We encourage you to dive deep into the literature and explore what other innovators are doing in the space you are interested in. Citing your sources where appropriate adds credibility to your ideas and perspective.

We further encourage you to ***think and research deeply before using AI*** to assist in your brainstorming. This will accelerate your learning, allow you to think more abstractly, and make your solution more unique and interesting.

The Prompt

Create a computational, ML-based, or other solution to improve…

- Vaccine uptake
- Pharmaceutical manufacturing accuracy and/or efficiency
- Computational pre-clinical drug development
- Tolerability, adherence, and outcomes for GLP-1 therapies
- Clinical trial recruitment

Vaccine Uptake

Questions to Consider

- What are the different drivers of low vaccine uptake?
- How can we address vaccine skepticism without being dismissive?
- What methods have been tried in the past? How did they succeed or fail?
- How do pharmaceutical companies, NGOs, and governments work together on vaccination campaigns? How might these paradigms change going forward?
- What population or disease type will you focus on? Or will your focus be more generic?
- What regulatory burden do public health campaigns fall under?
- How can we better predict vaccine uptake? How can this connect to public health recommendations and epidemiology forecasting?

Resources for Further Reading

- Understanding Vaccine Hesitancy: Insights and Improvement Strategies Drawn from a Multi-Study Review — PMC
- Internal and external factors affecting vaccination coverage: Modeling the interactions between vaccine hesitancy, accessibility, and mandates — PMC
- The Cutter Incident: How America's First Polio Vaccine Led to the Growing Vaccine Crisis — PMC
- 'Unqualified failure' in polio vaccine policy left thousands of kids paralyzed — Science | AAAS
- CGD-MillionsSaved_Case1.pdf
- 2023-2024 Case Studies — Sabin Vaccine Institute
- Real-world uptake of COVID-19 vaccination among individuals expressing vaccine hesitancy: A registry-linkage study — PMC
- Recipient-focused interventions to increase vaccine uptake in high and upper-middle income countries: a systematic review and network meta-analysis — eClinicalMedicine
- Forecasting the COVID-19 vaccine uptake rate: an infodemiological study in the US — PMC
- Interplay of epidemic spreading and vaccine uptake under complex social contagion — Phys. Rev. E

Manufacturing Accuracy and Efficiency

Questions to Consider

- What scale do you want to consider? Process specific? Drug class specific? A specific drug?
- How does manufacturing differ based on the type of drug (biologics, small molecules, vaccines)?
- How do we accurately forecast demand for drugs that must be used quickly to ensure adequate supply without overproduction?
- What bottlenecks exist in the manufacturing process at your chosen scale? What error types are common?
- Where can ML be integrated into a legacy process for forecasting risk, outcomes, or for automating a manual task?
- Are there hidden barriers to the augmentation you are interested in such as gaining trust from manufacturing operators, pausing production to integrate or test a new system?

Resources for Further Reading

- The Future of AI/ML in Pharmaceutical Manufacturing — Dassault Systèmes
- Pharmaceutical Company Increases Yield in Batch Manufacturing with AI Insights
- Regulatory Perspectives for AI/ML Implementation in Pharmaceutical GMP Environments — PMC
- A Comprehensive Strategic Analysis of Machine Learning Applications in the Pharmaceutical Industry — DrugPatentWatch
- The transformative power of artificial intelligence in pharmaceutical manufacturing: Enhancing efficiency, product quality, and safety — ScienceDirect
- A Quick-Start Guide to Biologics Manufacturing
- How Vaccines are Developed and Approved for Use — CDC
- Critical Components for Vaccine Manufacturing — Globally Resilient Supply Chains for Seasonal and Pandemic Influenza Vaccines — NCBI Bookshelf
- How AI and Machine Learning Can Bring Quality Improvements in Biopharma — AJMC
- Machine Learning for CMC Process Optimization: A Guide — IntuitionLabs

Computational Pre-Clinical Drug Development

This is an extremely broad topic. To help focus your scope, consider the following questions:

Questions to Consider

- What broad pharmaceutical category are you focused on (e.g., vaccines, biologics, small molecules, biomarker discovery)?
- Will your solution be focused on genomics, transcriptomics, proteomics, structural biology, or something else?
- Is your focus on finding new drugs, repurposing old drugs, refining and improving a drug candidate, predicting ADMET features, predicting the druggability of a target, identifying targets, or something else?
- What computational methods already exist in relation to your topic? How are you changing or improving them (e.g., by creating a larger workflow)?

Additionally, consider:

- What is the traditional lab-based method you are trying to augment?
- How will your method interface and iterate with this method?
- What data sets do you need to train or build your solution? Do they exist? Do they cost money? If existing datasets are too small, can you augment them?

Resources for Further Reading

- Structural Biology Basics: Techniques and Modeling — Technology Networks
- Computational Approaches in Preclinical Studies on Drug Discovery and Development — PMC
- From preclinical to clinic faster with mechanistic modeling and AI — Drug Discovery News
- AI-driven virtual cell models in preclinical research: technical pathways, validation mechanisms, and clinical translation potential — npj Digital Medicine
- Computational tools for ADMET
- Revisiting Protein–Protein Docking: A Systematic Evaluation Framework — Journal of Chemical Information and Modeling
- Large language models for drug discovery and development — ScienceDirect
- Open Targets Platform
- Rosetta Commons

Tolerability, Adherence, and Outcomes for GLP-1 Therapies

Questions to Narrow Your Focus

- Which axis of the problem are you targeting (tolerability, adherence, outcomes)?
- What is your patient population and indication?
- What point in the patient's journey are you targeting (initiation, titration, side-effect management, plateau, discontinuation, etc.)?
- What signals are you planning to use?
- How will your tool interface with pre-existing ecosystems (EHR, e-prescription software, PBM adherence programs)?
- What is your potential equity and access lens?
- Is there any regulatory framing for your digital component?

Resources for Further Reading

- Safety and Tolerability of Glucagon-Like Peptide-1 Receptor Agonists — Drugs, Dec 2025
- GLP-1 Receptor Agonists: Therapeutic Potential and Emerging Concerns — Current Pharmacology Reports, Apr 2026
- Comprehensive evaluation of GLP-1 receptor agonists: an umbrella review — Nature Communications, Jan 2026
- The expanding role of GLP-1 receptor agonists — eClinicalMedicine, Aug 2025
- Discontinuation and Reinitiation of Dual-Labeled GLP-1 RAs Among US Adults With Overweight or Obesity — JAMA Network Open, 2025
- Three-Year Real-World Adherence and Persistence to GLP-1 RAs — Prime Therapeutics, AMCP Nexus 2025
- Semaglutide in a real-world outpatient setting — Obesity and Endocrinology, Oct 2025
- Causes and consequences of discontinuation of GLP-1 RAs or tirzepatide — Nature Reviews Endocrinology, May 2026
- GLP-1 Adherence in the Real World: Why Patients Stop and How Clinicians Can Help — Diabetes In Control, Mar 2026
- Metabolic rebound after GLP-1 receptor agonist discontinuation — eClinicalMedicine, Jul 2025
- Effects of GLP-1 RAs After Treatment Cessation — J Gen Intern Med, Nov 2025
- What Happens When Patients Stop Taking GLP-1 Drugs? — Cleveland Clinic, Mar 2026

Clinical Trial Recruitment

Key Questions

- What are the typical recruitment goals and metrics for a study? How does this vary across drug and disease types?
- What are the typical challenges clinical trials face in recruitment?
- What are the key barriers to enrollment for interested patients? Which of them are based on a lack of trust?
- How does failing to meet certain recruitment metrics impact the usability of a study or applicability across diverse populations?

Clinical trial recruitment bears a heavy regulatory burden; be sure any proposed solution will be both ethical and legal.

Resources for Further Reading

- Better Recruiting, Better Care — Harvard Medical School
- Overview of clinical study designs — PMC
- Current Challenges in Clinical Trial Patient Recruitment and Enrollment (PDF)
- Recruitment Challenges in Clinical Trials for Different Diseases and Conditions — Public Engagement and Clinical Trials — NCBI Bookshelf
- Building trust in clinical research: a systems approach to ethical engagement and sustainable outcomes — PMC
- Recruitment Planning — Penn Medicine Clinical Research, Perelman School of Medicine at the University of Pennsylvania
- Why Diverse Representation in Clinical Research Matters — Improving Representation in Clinical Trials and Research — NCBI Bookshelf
- The Importance of Diversity in Clinical Trials — Pfizer
- FDA — Enhancing the Diversity of Clinical Trial Populations (PDF)
- Here’s What We Know: Diversity in Clinical Trials — Patient Care, Weill Cornell